

# GEOSCIENCE

## *A Teacher's Guide*

The goal of this teacher's guide is to assist you with getting the most out of our Geoscience course. Within the guide, you will find:

|                                                                                                                                                                                                                                      |                                                                         |
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*Disclaimer: Please keep in mind, especially with the science standards alignment, that we don't claim to be experts in the educational standards of your particular state or country. This guide is here to help you take the reins of your student's education - if you feel like you're inadvertently veering off course, relax, have fun, and jump back in.*

## Tips for Using the Notes

Each lesson in our Geoscience class contains a video and doodle-style notes to go along with it.

You'll find which pages go with each lesson in the table of contents.

There is also an answer key for the notes that contains answers in blue text. Both PDF files can be downloaded from the introduction page of the course website.

There are multiple approaches for using the notes. Choose the approach that works best for your learner!

Here are some ideas:

- **Study the notes before the lesson:** Become more familiar with the topic by reading over the selected pages. Fill in any labels that you already know. Think about what you expect to learn.

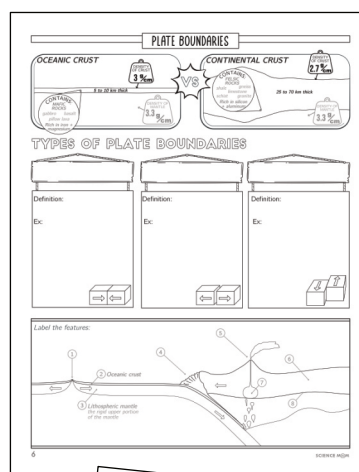
THEN

- **Fill out notes during the lesson:** pause the video and rewatch as needed.

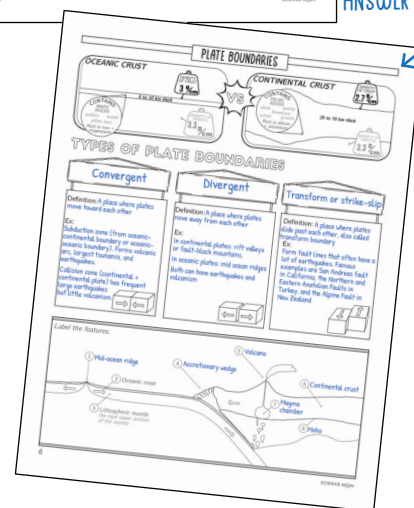
OR

- **After the lesson:** fill in the notes after watching as a way to synthesize and review what was learned. Complete them after class or the following day as a review. Refer to the video as needed.

We hope you find the notes to be a valuable tool for deeper learning and reinforcement of key concepts.



ANSWER KEY



## Time Commitment and Suggested Pacing

We suggest 3 lessons per week. Lecture-style lessons are about 45 minutes. The hands-on activities will likely take around an hour. For those adding up their hours, that's about 2.5 hours per week.

For a thorough student who enjoys extensions and chooses to explore the deep-dive resources with each lesson, plan 1.5 hours per lesson and 2 hours per hands-on activity, or approximately 5 hours per week.

### 5 lessons per week

*Most weeks will contain 2 hands-on activities.*

**Time to complete course:**

9 weeks

### 3 lessons per week

*Matches the pacing in the table of contents.*

*2 lecture-style lessons and 1 hands-on activity each week.*

**Time to complete course:**

15 weeks

### 2 lessons per week

*Will have a hands-on activity every other week.*

**Time to complete course:**

22 weeks

# Supply List for Hands-on Activities

## Lesson 3 - A Ring of Fire

- Pencil & colored pencils or crayons
- Internet connection or book(s) to use for researching volcanoes and earthquakes

## Lesson 6 - Shake, Rattle, Resilient

- A small box
- Cardboard (at least 3x as long as the small box)
- Cylindrical pencils or markers
- Smart phone
- Various household objects

## Lesson 9 - Stream Table Study

- Sand and gravel
- Plastic paint tray or a long bin or storage container
- Drill or nail
- Rocks or brick to elevate tray/container
- Cups or an empty gallon jug

## Lesson 12 - Geology Assessment

*Students could make the geology assessment into an activity by:*

- *Making a geology poster with key terms and concepts*
- *Creating a new set of questions and using that to quiz friends or family members*

## Lesson 15 - Cloud in a Jar

*Adult supervision recommended*

- 4 glass jars with lids
- Ice
- Water
- Matches
- Paper or tape and pen for making labels

## Lesson 18 - Humidity Lab

- A small piece of cloth or gauze
- Fan
- Rubber band
- 2 identical thermometers

## Lesson 21 - Convection Convention

*Adult supervision recommended*

- 2 to 4 identical clear cups
- 2 paper cups (will need to be cut)
- Stiff wire, taller than the paper cups
- Scissors or exacto knife
- Thin flat piece of plastic
- Food coloring
- Water (some of it heated to be very warm)
- Ice cubes
- Salt
- Tray
- 1 large clear container
- Pencil
- Matches
- Tea candle

## Lesson 24 - Tropical Storm Quest

- Pencil & colored pencils or crayons
- Internet connection or book(s) to use for researching a historic hurricane

## Lesson 27 - Weather and Atmosphere Assessment

*Students could make the atmosphere assessment into an activity by:*

- *Making a weather and atmosphere poster with key terms and concepts*
- *Creating a new set of questions and using that to quiz friends or family members*

## Lesson 30 - Stackable Food Chain & Food Web

- Cardboard
- Colored pencils, crayons, or markers
- Yarn
- Tacks or pins
- Scissors
- Gluestick

## Lesson 33 - Competing Compost Jars

- 2 identical clear containers
- Lids for the containers with ventilation holes OR 2 pieces of cloth and 2 rubber bands
- Newspaper
- Scissors
- Grass clippings or vegetable scraps such as carrot peels, apple cores, or squash rinds etc
- A small sample of soil, if possible, containing invertebrates such as earthworms, millipedes etc

## Lesson 36 - Invasive Species Comic

- Cardboard
- Colored pencils, crayons, or markers
- Scissors
- Gluestick

## Lesson 39 - Natural Resource Scavenger Hunt

- Pencil and lesson handout

## Lesson 42 - Ecology and Human Systems Assessment

*Students could make the ecology assessment into an activity by:*

- *Making an ecology and human systems poster with key terms and concepts*
- *Creating a new set of questions and using that to quiz friends or family members*

# UNIT ONE

## 1. Continental Drift (pages 4-5)

*Plate tectonics is the bedrock theory of geology. It explains most geological phenomena on our planet.*

### Vocabulary

|                                     |                |            |
|-------------------------------------|----------------|------------|
| Inner Core                          | Outer Core     | Mantle     |
| Continent                           | Tectonic Plate | Convection |
| Mohorovičić discontinuity or “Moho” |                |            |

### Learning Objectives

After this lesson, the student should be able to:

- Describe the theory of plate tectonics
- Explain that plate movement is influenced by convection in the mantle
- Identify and name the following major plates on a world map:
  - African Plate*                      *Antarctic Plate*
  - Australian Plate*                *Eurasian Plate*
  - Indian Plate*                      *North American Plate*
  - Pacific Plate*                      *South American Plate*
- List the main pieces of evidence that led to the discovery of plate motion (“puzzle piece” shape of continents, matching fossils and rock formations on different continents, mid-ocean ridges, magnetic striping of sea floor)

### Extensions

Woods Hole Oceanographic Institute articles about Marie Tharp:

An edited excerpt of “Connect the Dots: Mapping the Seafloor and Discovering the Mid-ocean Ridge” by Marie Tharp, from *Lamont-Doherty Earth Observatory of Columbia Twelve Perspectives on the First Fifty Years 1949-1999*

Marie Tharp: Pioneering Mapmaker of the Ocean Floor

Picture Books:

*Ocean Speaks: How Marie Tharp Revealed the Ocean’s Biggest Secret* by Jess Keating

Magnetic Striping and Polar Reversals:

USGS has a page in its FAQ section with [several good articles](#) relating to magnetic field reversal.

NASA has a good article called [“Flop Flop: Why Variations in Earth’s Magnetic Field Aren’t Causing Today’s Climate Change”](#)

### Conversation starters for the dining room table:

Was the scientific community correct in their initial rejection of Wegener’s theory? Why or why not?

Joke: Pangea was my favorite band. Earth just hasn't been the same since it broke up!

## 2. Plate Boundaries (pages 6-7)

*The movement of tectonic plates is the cause of many of Earth's geologic features, including mountains, rift valleys, and ocean basins. Tectonic plates are also responsible for most volcanoes and earthquakes.*

### Vocabulary

Convergent boundaries

Subduction

Rift Valley

Divergent boundaries

Continental Crust

Hotspot

Transform boundaries

Oceanic Crust

### Learning Objectives

After this lesson, the student should be able to:

- Explain how movement at the three types of plate boundaries causes earthquakes, volcanoes, and mountain building
- Identify convergent boundaries, including subduction and collisions, as places where plates come together
- Identify divergent boundaries, including rifts and mid-ocean ridges, as places where plates separate
- Explain transform boundaries as places where adjacent plates shear past each other

### Extensions

#### Conversation starters for the dining room table:

If you made a geology version of the rock, paper, scissors game, which boundary would defeat which? What hand signals would you make for convergent, divergent, and transform? Create your rules and then play a few rounds!

If you were a scientist studying Cotopaxi and Mauna Loa in Hawaii, what evidence would you look for to figure out whether these mountain were formed by a plate boundary or hotspot?

What tectonic plate do we live on?

Where is the nearest boundary? What type of boundary is it?

### 3. Activity: Ring of Fire (pages 8-11)

*The Pacific Ring of Fire is the most geologically active zone on Earth. It contains seventy five percent of Earth's active volcanoes and ninety percent of the world's earthquakes!*

#### Learning Objectives

After this activity, student should be able to:

- Explain why major volcanic eruptions and earthquakes are most often located near plate boundaries
- Describe how the severity of volcanic eruptions and earthquakes are measured: Volcanic Explosivity Index (VEI) and moment magnitude scale (Mw)
- Draw or locate the "Ring of Fire" on a world map
- Identify resources with additional information about the Ring of Fire, earthquakes, or volcanic eruptions.
- List tips for verifying if a resource is reliable or not (Who made it? What evidence does it use? Why was it published? Can the information be confirmed?)

#### Extensions

*Online resources:*

- The [National Geographic Ring of Fire](#) website has an [interactive mapmaker tool](#) where students can adjust layers and look at where plate boundaries, earthquakes, and volcanoes occur.
- USGS [Ring of Fire page](#) has several articles about the ring of fire and an earthquake map
- The [USGS Earthquake Hazards Program](#) has a map of latest earthquakes and a catalog where you can look up historic earthquakes
- [USGS](#) - the main page has links to the latest volcanic and earthquake notifications
- Smithsonian Institution National Museum of Natural History Global Volcanism Program [Global Volcanism Program | Current Eruptions](#)
- National Science Foundation's Seismological Facility for the Advancement of Geoscience ([SAGE](#)) has a seismic monitor page with a great visual of where recent earthquakes have occurred around the world. They also have a video series on geoscience careers, articles that explain how seismographs work, etc.

*Books:*

*Science Comics: Volcanoes: Fire and Life* by Jon Chad

*Fault Lines: Understanding the Power of Earthquakes* by Johanna Wagstaffe

*Volcano Atlas: An Epic Journey Around the World's Most Incredible Volcanoes* by Tom Jackson and Maggie Li

*DK Eyewitness Volcano and Earthquake*

*The Great Quake: How the Biggest Earthquake in North America Changed Our Understanding of the Planet* by Henry Fountain

*A Crack in the Edge of the World* by Simon Winchester

#### Conversation starters for the dining room table:

How can you tell if a source is reliable?

Why do subduction zones produce the world's largest earthquakes?

Joke: How should an excellent student of volcanology graduate? Magma cum laude.

Joke: How could you recognize the Ring of Fire when it was a toddler? It was always throwing a tantrum and making mountains out of molehills!

## 4. Weathering vs Tectonics (pages 12-13)

*Mountains are being raised up and weathered away in a never ending battle between tectonic forces and erosion.*

### Vocabulary

|                       |                     |                       |
|-----------------------|---------------------|-----------------------|
| Weathering            | Erosion             | Deposition            |
| Uplift                | Subduction          | Tectonic forces       |
| Mechanical weathering | Chemical weathering | Biological weathering |

### Learning Objectives

After this lesson, the student should be able to:

- Distinguish between tectonic forces (uplift, subduction, volcanism) and weathering/erosion processes
- Explain how tectonic forces can build and raise landforms while weathering and erosion wear them down
- Explain why famous mountain ranges such as the Himalayas and Appalachians are growing or shrinking from year to year
- Identify examples of mechanical weathering, chemical weathering, and biological weathering

### Extensions

Lichens have fascinating and diverse abilities for biological weathering. Learn more about them with these resources:

British Lichen Society webpage [What is a Lichen?](#)

Smithsonian Webcast [Exploring the Amazing World of Lichens](#) with Manuela Dal Forno

### Conversation starters for the dining room table:

How does the hardness of a material help you predict how it might change when exposed to the elements?

Imagine you could watch a mountain range for millions of years. What clues would help you figure out whether tectonic forces or weathering is “winning” in that location?

Joke: Geologists never get stressed. They just let things weather out.

Joke: Erosion is a crime - it's terrain robbery!

Joke: What did Sherlock Holmes say when Watson asked him what type of rock he'd found? "Sedimentary, my dear Watson!"

## 5. Faults and Earthquakes (pages 14-15)

*When rocks experience stress, they can bend or break.*

### Vocabulary

|                     |                     |                     |
|---------------------|---------------------|---------------------|
| Tension             | Compression         | Shear stress        |
| Deformation         | Strain              | Stress              |
| Brittle deformation | Ductile deformation | Elastic deformation |
| Normal fault        | Reverse fault       | Strike-slip fault   |
| Epicenter           | Focus               | Seismic wave        |

### Learning Objectives

After this lesson, the student should be able to:

- Identify the 3 major types of stresses on rocks (tension, compression, shear)
- Differentiate between brittle, ductile, and elastic deformations
- Describe the 3 major fault types and their associated movements
- Describe different seismic wave types and how they are measured

### Extensions

- Javalab Seismometer demo: [https://javalab.org/en/seismometer\\_and\\_inertia\\_en/](https://javalab.org/en/seismometer_and_inertia_en/)
- Javalab P-wave and S-wave demo: [https://javalab.org/en/seismic\\_wave\\_en/](https://javalab.org/en/seismic_wave_en/)
- The Pacific Northwest Seismic Network has a good page with information on the [Cascadia Subduction Zone Megathrust fault](#)
- [USGS](#) - scroll down to “real-time data” and pull up the map for earthquakes. Use the settings to toggle on faults. Then look at your location. Click on layers and turn on faults (available for US locations) and/or plate boundaries (available for all locations). Then open settings and show earthquakes from the past 30 days. How do earthquakes relate to faults and plate boundaries? Were there any earthquakes near your area? Did you feel them? Why or why not?

### Conversation starters for the dining room table:

What is the difference between an epicenter and a focus?

What clues from a fault or earthquake could help you figure out how much energy was released during the rupture?

Joke: Geologists aren't perfect - but all of their faults are stress related.



## 6. Activity: Shake, Rattle, Resilient (pages 16-17)

*There is no such thing as “earthquake-proof” but there are many ways to make structures more earthquake-resistant.*

### Learning Objectives

After this activity, student should be able to:

- Plan and carry out a fair experiment by changing just one variable between iterations
- Interpret quantitative data (e.g., acceleration readings) and use that information to evaluate which design worked best to minimize shaking
- Connect the results of the cardboard box experiment to the engineering designs used to minimize damage to larger buildings

### Extensions

*Online resources*

- Practical Engineering has a good article about [Tuned Mass Dampers in Skyscrapers](#).
- A paper by Koji Nakahara and others titled: [EARTHQUAKE RESPONSE OF ANCIENT FIVE-STORY PAGODA STRUCTURE OF HORYU-JI TEMPLE IN JAPAN](#)
- The NSF's Seismological Facility for the Advancement of Geoscience (SAGE) has an short animation of buildings on different types of rock experiencing different types of earthquake waves. It's called [Buildings & Bedrock: Effects of amplification](#).

### Conversation starters for the dining room table:

What is the nearest active fault to where you live? Has it ever had an earthquake?

Joke: Want a ground-breaking career? Be a seismologist!

Joke: What did the house say to the earthquake? You crack me up!

## 7. Go with the Flow (pages 18-19)

*Rivers are engines of erosion that transform the landscape.*

### Vocabulary

|            |                      |               |
|------------|----------------------|---------------|
| Stream     | Headwaters or Source | Tributary     |
| Oxbow Lake | River mouth          | Sediment load |
| Delta      | Alluvial fan         | Point bar     |
| Cut bank   | Continental divide   | Watershed     |

### Learning Objectives

After this lesson, the student should be able to:

- Define the term watershed
- Identify the parts of a stream system (eg headwaters, tributaries, main trunk, mouth) and identify which parts are associated with sediment production, transport, and deposition
- Identify and explain how deltas and alluvial fans form from river deposition
- Explain why rivers curve by describing variations in stream velocity at different depths and locations, and how these variations contribute to the formation of a point bar and cut bank, and the exaggeration of the streams meandering over time

### Extensions

- A NASA photo of the day called [Meandering Mississippi](#) is great for a discussion on state or country borders, which often use rivers. The image shows the current Mississippi River with a yellow line showing the border between Arkansas and Mississippi. What do you notice about the yellow line (which used to trace the river) and the modern river path?
- The USGS article [How Streamflow is Measured](#) has information about how scientists measure the amount of water in rivers. It's a complicated and interesting topic!

### Conversation starters for the dining room table:

If you were building a home next to the meander of a river, which side would you build a home on and why?

What clues help predict where erosion will happen and where new land will form around a river?

Joke: How does a river keep cool on hot summer days? With an alluvial fan.

Joke: What do you get when you cross a stream and a river? Wet feet.

## 8. Permafrost and The Cryosphere (pages 20-21)

*Most of the world's freshwater is frozen. Glaciers and icecaps are engines of erosion that move massive amounts of sediment.*

### Vocabulary

|                   |               |               |
|-------------------|---------------|---------------|
| Cryosphere        | Permafrost    | Glacier       |
| Crevasse          | Moraine       | Till          |
| Accumulation zone | Ablation zone | Albedo effect |

### Learning Objectives

After this lesson, the student should be able to:

- Define the cryosphere and identify its major components including land ice, sea ice, and permafrost
- Explain how glaciers are engines of erosion and shape landforms.
- Describe two major consequences of the cryosphere shrinking.

### Extensions

- Google Earth Engine Timelapse of [Columbia Glacier Retreat](#). This timelapse shows the retreat of the glacier from 1985 to 2022.
- A [virtual tour](#) of the Permafrost Tunnel at the Permafrost Tunnel Research Facility (PTRF) near Fairbanks, Alaska. The tunnel is rarely open to the public, so in 2023 they hosted a live virtual tour and the recording is an excellent way to learn more about permafrost in general as well as the tunnel and current research.
- This [VR Glaciers and Glaciated Landscapes](#) site put together by Dr. McDougall through the University of Worcester looks promising (disclaimer - I haven't explored it myself)

### BOOK RECOMMENDATION:

*The Two-mile Time Machine: Ice Cores, Abrupt Climate Change, and Our Future* by Richard B Alley

### Conversation starters for the dining room table:

Joke: What do you call a duck stuck in a glacier? A quack in the ice

Joke: Why do glaciers avoid warm hugs? They prefer to stay ice-o-lated

## 9. Activity: Stream Table Study (pages 22-23)

*Experience erosion firsthand by constructing a landscape and unleashing a river on it.*

### Learning Objectives

After this activity, student should be able to:

- Model how moving water shapes landforms using a stream table and connect their observations to real landforms such as meandering streams, deltas, alluvial fans, etc.
- Plan and conduct a scientific experiment by changing one variable and observing the impact on channel shape and sediment movement.

### Extensions

*Online resources:*

- If building a larger stream table, there are some tutorials or instruction pages that might be helpful:
  - [How to Build a Recirculating Stream Table](#), hosted by Texas A&M University
  - The American Geosciences Institute also has [instructions](#) for setting up and using a stream table.
- If your stream table got either too steep or too wet, you likely observed a **landslide** where a large portion of sediment moved all at once. The [U.S. Landslide Inventory and Susceptibility Map](#) by USGS has an interactive map where students can explore landslide risk by area and learn more about landslides.
- At <https://landslides.nasa.gov/> students can learn more about landslides and view NASA's [Daily Global Landslide Exposure Map](#). The map uses precipitation, soil moisture, and topography information to calculate landslide risk. In addition to the map, there is a [Global Viewer](#) where you can view real-time precipitation amounts around the globe.

### Conversation starters for the dining room table:

If the river of your stream table was a real river near a town, where would you build and where would you never build—and why?

Did any of your streams remind you of a real-world river system? If yes, what type of river or stream did they resemble? A steep mountain stream? Meandering river? Braided river?

Joke: Why is it a bad idea to invite a landslide to dinner? They take things downhill fast.

## 10. How Old are Rocks (pages 24-25)

*The techniques of stratigraphy and radiometric dating can be used to compare and find the ages of rocks.*

### Vocabulary

|                  |                    |                |
|------------------|--------------------|----------------|
| Fossil           | Mineralization     | Stratigraphy   |
| Relative dating  | Radiometric dating | Isotope        |
| Superposition    | Crosscutting       | Parent isotope |
| Daughter isotope |                    |                |

### Learning Objectives

After this lesson, the student should be able to:

- Explain the difference between relative dating and absolute dating of rock layers
- Describe the principles of stratigraphy such as superposition and crosscutting and explain how they can be used to determine the relative age of rock layers
- Define an isotope, and how ratios of isotopes can be used in radiometric dating

### Extensions

- The University of Colorado Boulder has a Phet simulation called [Radioactive Dating Game](#) which allows students to explore carbon dating and half life in more detail

### *Conversation starters for the dining room table:*

How does the idea of half-life help scientists estimate the age of rocks or fossils?

What would make some isotopes more useful than others for dating very old or very young materials?

Joke: What did the scientist say when he found 2 isotopes of helium? Hehe

Joke: Why did the rock cross the road? -To demonstrate the principle of lateral continuity!

## 11. Geologic Time (pages 26-27)

*The Earth has existed for billions of years. To better understand Earth's history, scientists have divided it into eons, eras, periods, and epochs. The supercontinent Pangea existed for approximately 100 million years before breaking up during the beginning of the Jurassic period.*

### Vocabulary

|                           |                |        |
|---------------------------|----------------|--------|
| Index fossil              | Supercontinent | Pangea |
| Stratigraphic correlation | Eon            | Era    |
| Period                    | Epoch          |        |

### Learning Objectives

After this lesson, the student should be able to:

- Define an index fossil and explain why they are useful for dating rock layers
- Describe stratigraphic correlation
- Recognize and name some of the eons, eras, and periods of the geologic time scale

### Extensions

- The [EarthViewer](#) page from hhmi (Howard Hughes Medical Institute) has an interactive view of Earth with a slider that lets students rewind time and see the movement of continents.
- Australia's National Science Agency has an interactive [chronostratigraphic chart](#) where students can explore the geologic timescale and see it with a logarithmic or linear scale, which helps show how long some of the early periods were.
- The University of Berkeley also has an interactive [geologic timescale](#) where each eon and era has links that tell more about the fossils and life of that time.
- The Dinosaur Database has an [interactive globe of ancient Earth](#) where students can select different milestones (such as first corals, land plants, or dinosaurs) and see what Earth looked like during that period.

### BOOK RECOMMENDATIONS:

- The Map that Changed the World by Simon Winchester
- The Ends of the World: Volcanic Apocalypses, Lethal Oceans, and Our Quest to Understand Earth's Past Extinctions by Peter Brannen

### Conversation starters for the dining room table:

If you could use a time machine to visit a previous geologic period, which would you choose to see and why?

What does "the present is the key to the past" mean?

Why is geology perhaps the only area of study where "a few million years" is considered brief.

Joke: What do you call a pirate that digs for fossils?- An Arrrrr-cheologist

## 12 and 13: Geology Assessment and Quiz Show (pages 29-31)

*We have covered a lot of ground! Here are some ideas to review and/or celebrate material learned, as well as information on the invasive species that Math Dad dressed up as for students to defeat during the quiz show.*

### Ideas for Studying and Review:

- Play charades using the vocabulary terms from class
- Play the “Flyswatter Game” with geology vocabulary:
  - Write 20 of the vocabulary words on index cards and place them on a wall or table
  - Give flyswatters to each competitor
  - Read the definition of the vocabulary term. The first player to swat the word with the flyswatter gets to keep the card. The player with the most vocabulary cards at the end wins.
- Make a geology-themed poster such as:
  - An artistic representation of key terms and concepts
  - A review poster listing each lesson title and 3 facts about each lesson
- Host a family quiz show where the student writes 20 questions from the lessons and then quizzes family members

### Ideas for celebrating the end of the unit:

- Make an edible treat to commemorate stratigraphy:
  - Math Dad’s raspberry jello cake was an adaptation of the Raspberry Jello Cake from Natasha’s Kitchen
  - The Joy-Filled Kitchen and House of Nash Eats both have recipes for layered rainbow jello
  - “Fossil dig” pudding cups (chocolate pudding with layers of buried gummy candies and/or crushed pretzles or cookies as sediment layers)
- Plan a field trip or activity to tie-in principles learned:
  - Visit the nearest Natural History Museum
  - Look up a geologic map of your area and then go on a hike or walk and see if you can find some of the rocks mentioned

### Invasive Plant Information

*Each boss Math Dad dresses up as during the quiz shows in this class is named for a different invasive species. To learn more about *Reynoutria japonica* or Japanese Knotweed, see:*

- Penn State Extension’s 3 minute video on [how to identify knotweed](#).
- King County Noxious Weed Control Program has a PDF on [invasive knotweeds](#) with information on how to identify and remove the plants.
- [USGS has a page](#) with basic information as well as a map showing where knotweed has been observed in the United states and Canada.

## 14. Relative Humidity (pages 32-34)

*At higher temperatures, water molecules have more energy and more of them will be in the air as water vapor. Relative humidity is the percent of actual water vapor in the air versus the total amount it could hold at saturation. The air temperature when air is at 100% humidity is called the dew point.*

### Vocabulary

|                   |                           |               |
|-------------------|---------------------------|---------------|
| Evaporation       | Condensation              | Precipitation |
| Absolute humidity | Relative humidity         | Dew point     |
| Water vapor       | Saturation vapor pressure |               |

### Learning Objectives

After this lesson, the student should be able to:

- Explain how temperature impacts evaporation and condensation.
- Define relative humidity and explain how it is different than absolute humidity
- Define the term dew point.
- Explain the basic forces that drive the formation of an afternoon thunderstorm

### Extensions

- NOAA has a page with a [Relative Humidity Calculator](#). You can input the temperature and dew point and it will calculate the relative humidity.
- The UK met office has a page called [Understanding Humidity](#) with a great 4-minute explainer video on humidity. It explains why hair can become more frizzy with humid weather and also shows how hygrometers work to measure humidity.

### *Conversation starters for the dining room table:*

What's is relative humidity? How is it different than absolute humidity?

Have you ever seen an afternoon thunderstorm?

Joke: What do you call humidity with no sense of direction? -Mist-placed

Joke: Did you hear about the water that evaporated? -It will be mist.



## 15. Activity: Cloud in a Jar (page 35)

*Model cloud formation with water, ice, glass jars, and matches.*

### Learning Objectives

After this lesson, the student should be able to:

- Explain the impact that small particles (aerosols) have on cloud formation
- Explain whether cloud formation is more likely to occur over warm or cold water

### Extensions

- Weatherstem has an interactive [Cloud Formation](#) page where students can adjust surface temperature and dewpoint to see where clouds could form. The cloud formation page is part of a longer [module](#) that walks the student through how clouds form, different types of clouds, and why only some clouds produce rain.
- NOAA has a page called [How Clouds Form](#) that talks more about cloud nuclei.
- The UK MetOffice also has a good [page on clouds](#) that covers what clouds are and how they form.
- To see truly incredible pictures of clouds and better appreciate their variety, check out the [Cloud Appreciation Society](#)'s picture gallery. Their cloud of the day and cloud of the month photos are always spectacular and often contain information about how the clouds formed.

### *Conversation starters for the dining room table:*

When did you last see thick low lying clouds? Do you remember what the weather conditions were on that day?

When did you last see a sky with no clouds or only high wispy clouds? Do you remember what the weather conditions were on that day?

What do you think it would feel like to be inside a cloud?

Joke: Where do clouds keep their money? -In a fog bank

Joke: Why did I failed my paper on condensation? -I missed the dew date.

## 16. Heat Index and Windchill (pages 36-37)

*The human-perceived temperature can feel quite different than the actual temperature. When air temperatures is near or above human body temperature, high humidity prevents evaporation of sweat. The heat index communicates how dangerous high temperatures at high humidity are. When air temperatures are cold, increased wind speed will remove body heat more quickly.*

### Vocabulary

Heat index or humidex

Wind chill

### Learning Objectives

After this lesson, the student should be able to:

- Define heat index and windchill
- Explain what “human perceived temperature” is and give scenarios of when this is useful
- Use critical thinking to analyze a heat index chart and apply the information

### Extensions

- Did you know that wool can keep you warm even when its wet? Wool fibers have an outer layer called a cuticle that repels water while the inside is porous and absorbs moisture. To learn more, search “why wool keeps you warm when wet” or “wool fiber structure.” The George Washington University [Textile Museum](#) has a page about different fibers with interesting information on how different fabrics can help regulate body temperature by retaining warmth (like wool).
- NOAA has a page called [Meteorological Conversions and Calculations](#) that has calculators for both wind chill and heat index.

### *Conversation starters for the dining room table:*

In cool climates, people are advised that “cotton kills” and told to wear different fabrics when doing outdoor activities. What makes different fabrics “wicking”? Why would that come in handy?

Do you think a temperature of 30°C (86°F) is comfortable or not? Have you ever experienced that temperature at very low humidity levels? Have you ever experienced that temperature at very high humidity levels?

Would you rather have a summer where the temperature is always higher than normal but the humidity is low, or a summer where the temperature is lower than normal, but the humidity is high? Why?

Joke: What is faster, heat or cold? - Heat, because you can catch a cold.

Joke: How do dairy farmers know when it's really hot outside? -Their cows start producing evaporated milk!

Joke: How do you keep warm in a cold room? -Go to the corner because it's always 90 degrees.

## 17. Air Masses and Fronts (pages 38-39)

*Wind blows from an area of higher pressure to an area of lower pressure. Air masses are large parcels of air with the same characteristics pressure, temperature, and humidity. When a cold mass of air overtakes a warmer mass of air (cold front), this is associated with storms and intense precipitation. When a warm mass of air overtakes a cold mass of air (warm front) it also produces rain but in a much more steady manner.*

### Vocabulary

|                     |              |              |
|---------------------|--------------|--------------|
| Barometric pressure | Maritime     | Continental  |
| Cumulonimbus        | Nimbostratus | Cirrostratus |

### Learning Objectives

After this lesson, the student should be able to:

- Identify the major types of air masses and their source regions
- Distinguish between cold fronts and warm fronts and identify characteristics of each
- Define jet streams.
- Interpret and read a map of predicted weather patterns.

### Extensions

- Create an "old school" weather forecast map using warm and cold front symbols and low or high pressure area symbols. Present it to your family, friends, or classmates and explain how to read or interpret it.
- There are more types of fronts than just "cold" and "warm." Boundaries between cold and warm air masses that aren't moving are called stationary fronts and an occluded front occurs when a cold front overtakes a warm front.

### *Conversation starters for the dining room table:*

If you got to be inside a cloud, which type of cloud would you choose and why?

If you saw a map with a cold front coming from the west and a warm front from the south, what type of weather event might happen when they meet?

Joke: I tried to make a barometer joke, but I couldn't. The pressure was too high.

Joke: Why did the lady go outside with her purse open? -She expected some change in the weather!

Joke: What is a gust of winds favorite color? -Blew.

## 18. Activity: Humidity Lab (pages 40-41)

*Make a homemade psychrometer (a wet and dry bulb) to measure relative humidity.*

### Learning Objectives

After this lesson, the student should be able to:

- Explain the relationship between temperature and the atmosphere's capacity to hold water
- Explain why the wet bulb temperature is always lower or equal to the dry bulb

### Extensions

- The National Weather Service (via NOAA) has a [relative humidity calculator](#) where you can adjust the temperature and dew point to calculate humidity.

### *Conversation starters for the dining room table:*

Have you ever used a swamp cooler before? If you were stranded in a desert climate, could you make a homemade swamp cooler?

Joke: Why was the humidity so good at math? -It knew how to multiply the discomfort!

## 19. Global Weather Patterns (pages 42-43)

*Uneven heating of Earth's surface and the Coriolis effect cause global weather patterns. The rainiest regions on Earth are along the intertropical convergence zone. Large slowly rotation cells of air explain the location of the jet streams and why most of the world's deserts are at 30° latitude.*

### Vocabulary

|                           |                                       |                        |
|---------------------------|---------------------------------------|------------------------|
| Coriolis effect           | Intertropical Convergence Zone (ITCZ) | Hadley cell            |
| Ferrel cell               | Polar cell                            | Westerlies             |
| Easterlies or Trade winds | Polar jet stream                      | Subtropical jet stream |

### Learning Objectives

After this lesson, the student should be able to:

- Identify the major global circulation cells (Hadley, Ferrel, Polar) and locate them on a world map.
- Describe the Coriolis Effect and how the Earth's rotation impacts it.
- Classify the prevailing wind belts (westerlies, easterlies or trade winds etc) and indicate their direction

### Extensions

- See the jet stream in action. NASA's Scientific Visualization Studio has a [short video](#) showing 1 month of data on the polar jet stream showing how dynamic it is.

### *Conversation starters for the dining room table:*

If Earth spun twice as fast, how do you think global weather patterns would change?

Joke: What do you call it when a meteorologist predicts wind speeds? -A best gust-imate.

Joke: What do you call it when coins fly through the jet stream? -The winds of change.

## 20. Ocean Currents (pages 44-45)

*Each of the major ocean basins contain a gyre, a large system of rotating surface currents driven by prevailing winds. The thermohaline circulation system functions as a giant conveyor belt of ocean currents. The gulf stream and Antarctic circumpolar currents are two significant parts of this larger system.*

### Vocabulary

|             |                                     |                          |
|-------------|-------------------------------------|--------------------------|
| Gyre        | Prevailing wind                     | Upwelling                |
| Gulf stream | Antarctic circumpolar current (ACC) | Thermohaline circulation |

### Learning Objectives

After this lesson, the student should be able to:

- Identify the major ocean gyres
- Explain how the Coriolis effect influences the direction of wind and ocean currents
- Define the thermohaline circulation system and explain which forces drive these currents

### Extensions

- Did you know that radioactive isotopes released during the nuclear testing of the Cold War provided a powerful and tracable marker to determine the speed of the thermohaline circulation system? Basically, only the upper levels of the ocean were contaminated in the 1950s. Certain isotopes such as tritium and caesium-137 are produced by nuclear reactors but have no other natural sources. The Woods Hole Oceanographic Institute has a good article called [\*Transient Tracers Track Ocean Climate Signals\*](#) which explains more about these isotopes and their role in helping scientists measure ocean currents.

### BOOK RECOMMENDATIONS:

- Tracking trash: flotsam, jetsam, and the science of ocean motion by Loree Griffin Burns  
*Part of the "Scientists in the Field" nonfiction book series published by Clarion books. Written for ages 10-12.*
- Diving Deep: Using Machines to Explore the Ocean by Michelle Cusolito and Nicole Wong  
*Picture book, for ages 5-8.*

### ***Conversation starters for the dining room table:***

Choose a location on a coast somewhere on Earth. If someone put a message in a bottle at that location and dropped it in the ocean, where would you expect it to travel?

If you were floating on a raft in the ocean, which would move you more: a sail that caught wind blowing across the surface or a deep-water "sail" that caught the movement of deep-water currents? Why

Joke: Why is the ocean always on time? -It likes to stay current.

How do high and low pressure systems greet each other? -A high says, "HI," and the low says, "HelLOW."

## 21. Activity: Convection Convention (pages 46-49)

*Conduct four investigations to learn more about convection and how temperature and density impact the movement of fluids.*

### Learning Objectives

After this lesson, the student should be able to:

- Describe what a convection current looks like and explain what drives its motion
- Connect observations from these smaller demonstrations to large-scale circulation patterns

### Extensions

- Look up the haloclines in the cenotes and caves of Central America. BBC Earth Science has a short video called [Diving through a Natural Phenomenon](#). It's truly impressive how stable and visible a halocline can be!

### *Conversation starters for the dining room table:*

Question: If a large amount of freshwater was added to the ocean in an area with cold sinking salt water, would that ocean current be changed? If yes, how? Would it speed up or slow down?

What do you call a river made entirely from chocolate and sweeties?

-A confection current.

## 22. Ocean Currents (pages 50-51)

*ENSO (El Niño Southern Oscillation) is a change in sea surface temperature and trade winds in the Pacific Ocean. It impacts weather conditions around the entire world!*

### Vocabulary

|         |             |                 |
|---------|-------------|-----------------|
| El Niño | La Niña     | Niño 3.4 Region |
| ENSO    | Thermocline | Upwelling       |

### Learning Objectives

After this lesson, the student should be able to:

- Identify the key features of El Nino, La Nina, and Neutral phases
- Explain what an El Niño or La Niña watch means and how far in advance it is issued
- Describe the regional weather impacts of El Niño and La Niña on at least three world regions

### Extensions

- NOAA ENSO Blog has regular updates on current El Niño/La Niña conditions at [climate.gov/enso](https://climate.gov/enso)
- NASA Global Precipitation Measurement. Satellite data showing real-time rainfall anomalies linked to ENSO at [gpm.nasa.gov](https://gpm.nasa.gov)

### Conversation starters for the dining room table:

If you were taking a world-wide vacation, which ENSO phase would you choose and why?

Peruvian fishermen named El Niño centuries ago because the water sometimes became warm and fish disappeared around Christmas. They noticed a pattern and named it long before scientists could explain why it happened. What other natural patterns do you think humans have observed and named before understanding the cause?



## 23. The Weather Forecast (pages 52-53)

*In modern forecasting, meteorologists gather data from balloons, satellites, and radar. They run dozens of computer simulations to find the most likely predictions for future conditions.*

### Vocabulary

|            |                      |                     |
|------------|----------------------|---------------------|
| Radiosonde | Ensemble forecasting | Cone of uncertainty |
| Microscale | Mesoscale            | Synoptic scale      |

### Learning Objectives

After this lesson, the student should be able to:

- Identify why weather forecasts become less accurate further into the future
- Describe the main data-collecting tools used in weather forecasting measure and what they measure (ex radar, satellites, weather balloons or radiosondes)
- Correctly interpret what a percent chance of rain means
- Explain why running multiple models with slightly different starting conditions (ensemble forecasting) improves the accuracy of the weather forecast

### Extensions

- Read your local weather forecast discussion and then explain it to a family member or friend. Bonus activity: translate it into slang speak or Victorian English.
- Check out the weather forecast (updated daily) at <https://www.weather.gov/>
- View live data from doppler radar to see where it's currently raining in North America at <https://radar.weather.gov/>

### *Conversation starters for the dining room table:*

The butterfly effect says that a tiny difference in starting conditions can eventually produce a completely different outcome (ie, the flap of a butterfly wings could cause a disturbance that eventually produces a massive storm). If that's true, will weather forecasts ever be accurate beyond 14 days — or is there a hard limit set by physics?

What do you think is the most important weather forecasting tool? Doppler, weather stations, satellites, or radiosondes? Why do you think it's the most important?

Why was the weather so unpredictable? Because it had cloudy judgment.

Knock-knock. Who's there? Accordion. Accordion who? Accordion to the weatherman, it's going to rain tomorrow.

What does a meteorologist cry before he tees off? -Fore-cast!

## 24. Activity: Tropical Storm Quest (pages 54-57)

*Learn about tropical cyclones (also called typhoons or hurricanes) by studying the tracks of several famous storms. Research one large storm and write a report on its impact.*

### Learning Objectives

After this lesson, the student should be able to:

- Explain how sea surface temperature influences tropical cyclone formation
- Explain why tropical cyclones almost never form south of the equator in the Atlantic Ocean
- Present the impacts of one historical storm and explain how a community could prepare for a similar storm

### Extensions

- View current tropical storm and/or cyclone activity at the US Naval Research Laboratory's [Tropical Cyclone Web](#)
- View any historical hurricane path using NOAA's [Historical Hurricane Tracks](#) website.

### BOOK RECOMMENDATIONS:

- Isaac's Storm by Erik Larson

*An award-winning nonfiction account of the 1900 Galveston hurricane, which remains the deadliest natural disaster in American history. General nonfiction (adult audience).*

- A Furious Sky: The Five-Hundred-Year History of America's Hurricanes by Eric Jay Dolin

*This award-winning nonfiction account looks at how hurricanes in the Atlantic have shaped North American history and how far-ranging the impacts of these megastorms are and will continue to be. General nonfiction (adult audience).*

### Conversation starters for the dining room table:

The year is 1800 and you get to choose where the 3 biggest cities along the gulf coast will be built. Where would you build them and why? (Can choose another area such as Florida or the Philippines)

Riddle: What has an eye but cannot see? A hurricane.

## 25. Tropical Cyclones (pages 58-59)

*Tropical cyclones are called by different names depending on where they form. But typhoons, hurricanes, and cyclones are all the same type of storm: they form over warm ocean water, have a low pressure center, and can be incredibly destructive. Their paths and shape are strongly influenced by the Coriolis effect.*

### Vocabulary

|             |            |                  |
|-------------|------------|------------------|
| Storm surge | Wind shear | Tropical cyclone |
| Typhoon     | Hurricane  | Heat engine      |

### Learning Objectives

After this lesson, the student should be able to:

- Explain what conditions are needed for tropical cyclones to form.
- Identify and define the key features and impacts of a tropical cyclone such as the eye, rain bands, and storm surge

### Extensions

- NOAA has a website dedicated to explaining the phenomenon of storm surge with several good animations at <https://www.nhc.noaa.gov/surge/>
- Use the illustrations of the Saffir-Simpson wind scale ratings on page 59 of the notes to create scenes that show a hurricane's intensity. Use building blocks, modeling clay, or drawings.
- Research cold core vs. warm core cyclones

### *Conversation starters for the dining room table:*

What is usually the most damaging feature of a tropical cyclone?

Knock, Knock. Who's there? Augusta. Augusta who? Augusta wind 74 mph or higher could be the start of a hurricane

How do cyclones keep fit? They do a lot of rotations.

## 26 and 27: Weather and Atmosphere Assessment and Quiz Show (pages 60-63)

*We have covered a lot of material! Here are some ideas to review and/or celebrate material learned.*

### **Ideas for Studying and Review:**

- Play charades using the vocabulary terms from class
- Play the “Flyswatter Game” with weather and atmosphere vocabulary:
  - Write 20 of the vocabulary words on index cards and place them on a wall or table
  - Give flyswatters to each competitor
  - Read the definition of the vocabulary term. The first player to swat the word with the flyswatter gets to keep the card. The player with the most vocabulary cards at the end wins.
- Make a weather-themed poster such as:
  - An artistic representation of your favorite terms and concepts
  - A review poster listing each lesson title and 3 facts about each lesson
- Host a family quiz show where the student writes 20 questions from the lessons and then quizzes family members

### **Ideas for celebrating the end of the unit:**

- Make a craft to commemorate learning about weather:
  - A map of modeling clay showing the topography of your area and common weather patterns
  - Blue gelatin and whipped cream in clear glasses can create a fun “clouds in the sky” desert.
  - Make a hurricane roll cake (the cross-sections look like the spiral pattern of a cyclone)
- Plan a field trip or activity to tie-in principles learned:
  - Visit the nearest Natural History Museum
  - Look up the nearest weather station office to where you are. Read their most recent weather report. Some offices are staffed 24/7 so they have the ability to issue an emergency alert. In these cases, if you have a weather-related question you can call the office late in the evening. The meteorologist on call will often be available and happy to answer questions.
  - Go radiosonde hunting. The device from high altitude weather balloons – the sonde – eventually drifts back down to Earth. People go out to find and collect these as a hobby. It’s a bit similar to geocaching but instead of following GPS points and/or clues left by a person, people use the radio signal of the device to find it.

## 28. Understanding Ecosystems (pages 64-65)

*The living members of an ecosystem can be studied according to their food and carbon source. Primary producers form the base of an ecological pyramid and apex predators are at the top. The total amount of biomass produced in an ecosystem is referred to as its primary productivity.*

### Vocabulary

|                |                                |                              |
|----------------|--------------------------------|------------------------------|
| Biomass        | Trophic level                  | Producer                     |
| Consumer       | Decomposer                     | Detritivore                  |
| Apex predator  | Gross primary production (GPP) | Net primary production (NPP) |
| Photosynthesis | Respiration                    | Ecosystem                    |

### Learning Objectives

After this lesson, the student should be able to:

- Describe the roles of producers, consumers, and decomposers in an ecosystem and give examples of each
- Define trophic levels and classify organisms as primary, secondary, or tertiary consumers
- Illustrate how energy and carbon flow in a closed ecosystem
- Explain why energy decreases at each trophic level and why apex predators are always rare

### Extensions

- NASA has several images and videos that show how ecological productivity changes between different areas and seasons. Search for maps of global net primary productivity. In 2024, there was a study looking at measuring productivity on land using satellite data. Read the study and take a look at understanding a scientific paper OR look at the animation in Science News that shows photosynthesis levels changing in North America from March to November:
  - The study: [Determining Terrestrial Ecosystem Gross Primary Productivity From PACE OCI](#)
  - The news article: [NASA Scientists Map Plant Productivity with Data from Ocean Satellite](#)
- Watch *Life in the Undergrowth*, a BBC documentary with David Attenborough. This series highlights decomposers and detritivores, which are an essential part of every trophic level.

### Conversation starters for the dining room table:

If only 10% of energy passes from one trophic level to the next, how many pounds of plants would a lion need to eat (indirectly) to gain one pound of body weight?

If all the decomposers disappeared tomorrow, what would happen to an ecosystem?

What did the primary producer say to the consumer? - "Leaf me alone, I'm making my own food!"

How does a producer do its math? -By using Algae-bra!

## 29. Keystone Species (pages 66-67)

*Ecosystems are complex systems with many parts. A keystone species is an organism that has an especially strong influence on an ecosystem.*

### Vocabulary

|             |                  |                    |
|-------------|------------------|--------------------|
| Monoculture | Keystone species | Biodiversity       |
| Predator    | Mutualist        | Ecosystem engineer |
| Symbiosis   | Trophic cascade  |                    |

### Learning Objectives

After this lesson, the student should be able to:

- Explain keystone species hypothesis and give an example of how the removal of one species can cause dramatic ecosystem-wide change
- Describe a trophic cascade using the wolves-in-Yellowstone example
- Describe a keystone species that is not a predator (a mutualist or ecosystem engineer)

### Extensions

- English journalist George Monbiot's short 5 min documentary [How Wolves Change Rivers](#) has good visuals and is a striking summary of what we talked about with Yellowstone and the reintroduction of wolves.

### *Conversation starters for the dining room table:*

Do you think scientists should keep using the term keystone species, refine it, or come up with a new term?

Robert Paine removed sea stars from some tide pools and left others alone (control group). Why was having a control group important? Would his conclusion have been as strong without the control group?

Did you know that some species of hermit crabs have symbiotic relationships with sea anemones? The crabs let the anemones ride around on their shells and will even share food with them. It's a prime example of keeping your friends close and your anemones closer.

How do you know when a keystone species was at a party? The whole place fell apart after it left.

### 30. Activity: Stackable Food Chain and Food Web (pages 68-71)

*Explore ecosystems further by making your own stackable food chain and filling in a kelp forest food web.*

#### Learning Objectives

After this lesson, the student should be able to:

- Create a food web for a kelp forest or ecosystem of your choice and label each part using vocabulary terms such as producer, primary consumer, secondary consumer, and apex predator.
- Describe a simple food chain

#### Extensions

- Look up the classic nursery rhyme “There was an old lady who swallowed a fly.” Do you think the song teaches the concept of a food chain? Why or why not?
- Is there anything in your kelp forest food web that seems redundant? If you had to remove one species with the goal of causing as little change as possible to the rest of the system, which species would you choose and why?

#### *Conversation starters for the dining room table:*

The meal you're eating right now is part of a food chain. Can you trace it back to a producer? How many trophic levels are on your plate tonight?

### 31. Nutrient Cycles (pages 72-73)

*Carbon, nitrogen, and phosphorus are essential building blocks of living things. The amounts of each element remain relatively constant, but where they are in the ecosystem changes over time. Each element or nutrient has different reservoirs or places where it exists. Weathering, temperature, and living things all influence how these elements move through ecosystems.*

#### Vocabulary

|                   |                    |                  |
|-------------------|--------------------|------------------|
| Reservoir         | Inorganic nitrogen | Organic nitrogen |
| Limiting nutrient | Nitrogen fixation  | Sedimentation    |
| Weathering        | Hypoxic            |                  |

#### Learning Objectives

After this lesson, the student should be able to:

- Describe the main carbon pools (rock, soil, ocean, atmosphere, living things) and how carbon moves from one pool to another
- Explain the difference between inorganic and organic nitrogen and list two ways it can change from one form to the other
- Explain one way the phosphorus cycle differs from the nitrogen and carbon cycles
- Describe a limiting nutrient and explain its impact in an ecosystem.

#### Extensions

##### BOOK RECOMMENDATIONS:

- Know Soil, Know Life by David L. Lindbo, Deb A. Kozlowski, and Clay Robinson

This soil textbook is an great and approachable resource to learn more about soil science.

#### *Conversation starters for the dining room table:*

The total amount of carbon on Earth doesn't change, but the location of that carbon does change. Where did most of the carbon in your body come from? (Hint: you are what you eat!) Where do you think most of the carbon in your body has been in the last million years?

How do plants feel about the carbon cycle? -They think it's a breath of fresh air.

What happens to nitrogen when you put it in direct sunlight? -It becomes day-trogen.



## 32. Succession (pages 72-73)

*Succession describes both how ecosystems get established in the first place (primary succession) and how they recover after significant disturbances (secondary succession). Pioneer species and seed banks are important aspects of succession.*

### Vocabulary

|                    |                      |             |
|--------------------|----------------------|-------------|
| Primary succession | Secondary succession | Disturbance |
| Pioneer species    | Seed bank            | Muskeg      |
| Understory         |                      |             |

### Learning Objectives

After this lesson, the student should be able to:

- List examples of primary and secondary succession
- Define pioneer species and explain why they are often the first to grow in an area
- Give at least two examples of disturbances that cause secondary succession

### Extensions

- Look up Surtsey Island, a new island off the coast of Iceland. It first emerged above the ocean in 1963 and scientists have documented how primary succession on that island is taking place.

### *Conversation starters for the dining room table:*

An ecosystem is paved over to make a parking lot and then that parking lot is abandoned. Pick a location and then describe how the ecosystem would get re-established. Would this be an example of primary or secondary succession?

What did the barren, rocky soil say about the pioneer species? -There isn't a single thing we 'lichen' about it, but it's really starting to grow on us!

### 33. Choose Your Own Activity: Composting, Ecosystem Triptych, or Nutrient Passport (pages 76-79)

*Explore ecosystems further by making your own stackable food chain and filling in a kelp forest food web.*

#### Learning Objectives

After this lesson, the student should be able to:

- Create a food web for a kelp forest or ecosystem of your choice and label each part using vocabulary terms such as producer, primary consumer, secondary consumer, and apex predator.
- Describe a simple food chain

#### Extensions

- Look up the classic nursery rhyme "There was an old lady who swallowed a fly." Do you think the song teaches the concept of a food chain? Why or why not?
- Is there anything in your kelp forest food web that seems redundant? If you had to remove one species with the goal of causing as little change as possible to the rest of the system, which species would you choose and why?

#### *Conversation starters for the dining room table:*

The meal you're eating right now is part of a food chain. Can you trace it back to a producer? How many trophic levels are on your plate tonight?